Model Development Report - Part 3

Production | An installable GitHub research tool

1. Product and intended user

An installable, versioned Python repository for students/researchers reproducing EVS life-satisfaction analysis or predicting an omitted A170 response for an EVS-style respondent. Version 0.1.0 includes the frozen boosting model and its complete preprocessing. There is no hosted service or deployment. Repository: github.com/sknefi/joint-life-satisfaction-analysis.

2. Install and run

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lock.txt
python -m pip install --no-deps .
evs-predict --input examples/respondents.csv --output predictions.csv
python -m unittest discover -s tests -p 'test_*.py'
```

The model artifact lives in models/; default paths assume the repository root. Elsewhere pass --model /path/to/evs_v0_1_0.joblib. The notebook notebooks/quickstart.ipynb demonstrates synthetic profiles without requiring the respondent dataset. Source data access instructions and immutable-source rules are in data/README.md; make data, train, inference and develop-reports reproduce the analysis locally.

3. Inputs, outputs and scope

Input CSV columns: X001, X003, X025R, X028, X047E_EVS5, A009, A165, A173, cntry_AN. Use official survey category codes; blank values and -1..-5 become missing. Age must be 18-82, with 82 interpreted as 82+. Country is mandatory and must be one of the 30 training countries. Output: predicted_score (distribution median), probability_high_7_10 and p_1 through p_10. The predictor rejects A170, identifiers, weights, unexpected columns and invalid categories; writes are atomic.

4. Verification as a reusable product

Check	Observed result
Package and regression tests	17 tests passed; existing profiler checks retained. Clean installed-wheel check: True; executed notebook: True.
Artifact parity	120 synthetic cases; maximum probability difference after save/load and CLI round-trip: 0. Class order and probability sums checked.
Bad inputs and leakage	Tests cover missing codes, malformed/range/category errors, unknown countries, target exclusion, fold-local preprocessing and country isolation.
Independent production validation	Frozen model: n=5,007, MAE=1.213, log loss=1.711, RPS=0.0963. Not used to tune or select the model.
Automation and reproducibility	GitHub Actions installs the package and runs synthetic-data tests; recorded versions, seeds, hashes and saved preprocessing. Actual hosted CI status is documented separately.

5. Honest interpretation and maintenance

The probabilities are model estimates for historical EVS-style contexts, not validated individual confidence intervals or causal effects. No interval coverage is claimed. Production reliability bins are saved; final same-country and six-country prediction tests remain for Assure. New countries, survey instruments or later periods need external validation. Monitor input/missingness drift and error by country when new labelled data arrive; retrain and version only after that evaluation.

EVS/WVS (2024), Joint EVS/WVS 2017-2022 Dataset, ZA7505 v5.0.0, doi:10.4232/1.14320. Raw respondent data, local split IDs and secrets are excluded from distribution. This repository publishes aggregate results and a trusted project model; its instructions do not require redistributing the source survey.